MMI 806 · PRINCIPLES OF RESPONSIBLE AI

MSc in Mathematical Innovation · Open University of Kenya

Module 5: AI and Bias

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LEARNING OUTCOMES

1. Define bias in AI systems and identify the principal types of algorithmic bias.
2. Explain the pathways through which biased data or design decisions produce unfair AI outcomes.
3. Analyse a case study to trace the origins, propagation, and effects of bias from data to decision.
4. Evaluate and propose concrete, evidence-based steps to reduce bias in AI system design.

1 Module Overview

Bias in AI is not an accident or a malfunction. It is a predictable consequence of building systems that learn from data generated by an unequal world and making design choices that reflect the assumptions and blind spots of their creators. This module develops a rigorous understanding of what bias is, how it enters AI systems, and why its consequences fall most heavily on those already marginalised.

BIAS IS NOT ALWAYS INTENTIONAL

The most dangerous forms of AI bias are not produced by engineers who intend to discriminate. They are produced by engineers who do not notice their assumptions, by organisations that collect biased data without questioning it, and by systems that optimise for the wrong metric. Recognising that bias can be unintentional does not reduce the moral responsibility to prevent it — it increases it.

2 Defining Bias in AI Systems

DEFINITION: Algorithmic Bias

Algorithmic bias refers to systematic, unjustifiable disparities in the outputs or outcomes of an AI system, particularly where those disparities disadvantage groups defined by protected characteristics (race, gender, age, disability, religion, nationality, socioeconomic status) or historically marginalised populations.

> A bias is *systematic* when it is not random — it consistently disadvantages the same group. It is *unjustifiable* when it cannot be explained by legitimate, relevant factors.

3 A Taxonomy of AI Bias

3.1 Historical Bias

Bias embedded in training data that reflects historical discrimination or inequality in the real world.

Example: A word embedding model trained on historical text learns that "doctor" is associated with men and "nurse" with women — because historical texts reflect a period when these were empirical correlations rooted in discrimination.

3.2 Representation Bias

Training data that underrepresents certain groups, causing the model to perform poorly for those groups.

Example: Facial recognition systems trained predominantly on lighter-skinned faces misidentify darker-skinned individuals at much higher rates. Joy Buolamwini's Gender Shades study (2018) showed commercial systems had error rates up to 34% higher for darker-skinned women than lighter-skinned men.

3.3 Measurement Bias

The measurement instrument produces different quality data for different groups.

Example: Pulse oximeters (devices measuring blood oxygen level) are less accurate for darker-skinned patients. AI diagnostic systems trained on pulse oximeter data inherit this measurement error.

3.4 Aggregation Bias

Building a single model for a diverse population when subgroups have fundamentally different patterns.

Example: A diabetes risk model trained on predominantly white, Western populations may produce systematically different (and less accurate) risk estimates for East African populations with different genetic, dietary, and environmental profiles.

3.5 Deployment Bias

A system performs well in testing but is deployed in a context that differs from the training environment.

Example: A credit scoring model trained on urban Kenyan borrowers is deployed nationwide, including in rural populations with different financial behaviour patterns. Predictions for rural borrowers are systematically less accurate.

3.6 Feedback Loop Bias

AI-driven decisions create patterns in future data that confirm the original bias.

Example: A predictive policing algorithm directs officers to patrol certain neighbourhoods more heavily. More arrests occur there (because police are there). Future training data shows more crime in those areas — reinforcing the algorithm's predictions. The algorithm "proves itself right" by creating the patterns it predicted.

EXAMPLE 1: Amazon's AI Hiring Tool

What happened: Amazon built an AI recruitment tool to screen CVs and predict candidate quality. It was trained on 10 years of historical CVs submitted to Amazon — a period when the technology sector was (and remains) male-dominated.

The bias: The model learned to penalise CVs that included the word "women's" (as in "women's chess club" or "women's coding group") and to down-rank graduates of all-women's colleges. The model had learned from data generated by a biased hiring culture.

Amazon's response: Amazon disbanded the tool in 2018.

Bias type: Historical bias (training data reflected a biased hiring history) + representation bias (women underrepresented in historical hires).

4 Why Bias Disproportionately Harms Marginalised Groups

THE HARM AMPLIFICATION MECHANISM

AI bias is not merely inequitable — it is actively harmful to marginalised groups through three mechanisms:

1. **Systematic disadvantage:** Every interaction with the biased system produces a worse outcome for the disadvantaged group. This accumulates across millions of interactions.
2. **Obscured accountability:** Algorithmic discrimination is harder to challenge than human discrimination because the mechanism is opaque and the decision appears "objective."
3. **Authority effect:** AI outputs are often accorded higher credibility than human judgments, making biased outcomes harder to contest even when they are clearly wrong.

5 Designing a Fair AI System: A Worked Exercise

MODULE 5 ASSIGNMENT (max 500 words)

Scenario: You are designing an AI system to identify secondary school students who may need academic support in Kenya. The system will analyse student data and generate a "support needed" flag for teachers to follow up.

Reflect on:

1. **What data to include:** List at least six variables you might use and explain why each might be predictive.
2. **What data introduces bias:** For each variable, identify if and how it could introduce bias. Which variables would you exclude, and why?
3. **Representation:** How would you ensure that your training data represents students from all counties, income levels, genders, and disabilities?
4. **Fairness definition:** Choose one definition of fairness (equal error rates, calibration, or individual fairness) and explain why it is most appropriate for this context.
5. **Feedback loops:** How would you prevent the system from creating a feedback loop where flagged students receive less attention, perform worse, and are flagged again?

6 Key Concepts and Terminology

Algorithmic Bias	Systematic unjustified disparities in AI outputs disadvantaging certain groups.
Historical Bias	Bias from training data reflecting past discrimination or inequality.
Representation Bias	Bias from training data underrepresenting certain groups.
Feedback Loop	Cycle where AI decisions generate data that reinforces initial bias.
Gender Shades	Buolamwini & Gebru (2018) study showing facial recognition disparities by skin tone and gender.
Protected Characteristic	Attribute (race, gender, age, etc.) that must not be the basis for unjustified discrimination.
Word Embedding	Mathematical representation of words as vectors capturing semantic relationships; inherits bias from text.

7 Revision Questions

1. Classify each of the following as a type of bias: (a) a loan model performing worse for women because it was trained on data from a period when women had less access to formal credit; (b) a disease prediction model performing worse for South Asian populations because they were underrepresented in clinical trials; (c) a predictive policing tool used in a city different from where it was trained.
2. Explain the feedback loop bias mechanism in predictive policing using a mathematical example. If area A has twice as many officers as area B, and arrest probability is proportional to police presence, what happens to the risk scores for areas A and B over successive retraining cycles?
3. Amazon's recruitment AI was disbanded. Was this the right decision? Propose an alternative approach that would have addressed the bias without abandoning the tool.
4. Design a "bias audit" protocol for an AI system used to allocate bursaries to university students in Kenya. Describe what data you would collect, what metrics you would measure, and what thresholds would trigger remediation.
5. Explain measurement bias using the pulse oximeter example. How should this bias be addressed: (a) in the AI model; (b) in the data collection protocol; (c) in the clinical deployment?
6. Is it possible to build a completely unbiased AI? Argue your position using at least two of the bias types defined in this module.
7. A teacher tells you: "The AI support tool flags mostly students from low-income families. That's not bias — those students genuinely need more support." Critically evaluate this statement.

MODULE 5 SUMMARY

- AI bias is systematic, predictable, and often unintentional — arising from historical inequality in data, underrepresentation, measurement error, aggregation failures, deployment mismatch, and feedback loops.
- The six bias types (historical, representation, measurement, aggregation, deployment, feedback loop) require different interventions — recognising the type is the first step toward remediation.
- Bias disproportionately harms marginalised groups through accumulated disadvantage, obscured accountability, and authority effects.
- Designing for fairness requires explicit choices about which definition of fairness to optimise — choices that are ethical and political, not merely technical.
- *AI trained on an unequal world reproduces and amplifies that inequality unless designers actively intervene.*

8 Further Reading

- **Buolamwini, J. & Gebru, T.** (2018). Gender Shades. *Proceedings of Machine Learning Research*, 81.

- **Barocas, S., Hardt, M., & Narayanan, A.** (2023). *Fairness and Machine Learning*. MIT Press. Free at https://fairmlbook.org
- **Dastin, J.** (2018). Amazon scraps secret AI recruiting tool that showed bias against women. *Reuters*.